DroneVPS — Project Status

Marker-free GNSS-denied navigation on a 4.0 kg hexacopter. Target: 100-150 m AGL, 3-5 km out-and-back, no GNSS, no ground marker. Twenty-six flights across 19-25 October. Flight 245 is the first that worked: 634 seconds on the VPS, LOITER throughout, zero failsafe, and 0.44 m of position error at 38 m. The system is now limited by the map rather than by drift, which is a regime it has never been in before.

The phantom velocity was a model error, and it is gone

Derotation subtracted ONE centre-valued pixel shift from a 77.7 degree frame. A rotation moves a point at radius r by f·theta·(1 + r²/f²) — +25% at the top and bottom edge, +65% at the left and right — and the tracker reduces ~400 tracks spread across the whole frame to one number. Predicted shortfall over uniformly spread points: 25.6%. Residual measured on two flights by fitting step_px = a·rotation_px + b: a = 0.157 and 0.317. The rotation now comes off every track before the fit. That is why the coefficient sweep moved nothing (0.026 to 0.024) and the integration-window change was bit-identical: both were hunting a gain error inside a model error.

The camera is not looking straight down, and nothing said so

A tilt of theta puts the point under the middle of the frame alt·tan(theta) from the point under the aircraft, and the matcher reported that centre as the position. At 110 m a 10 degree tilt is 19.4 m. corr(tilt, fix error) +0.484. All four sign pairs were tried against the raw receiver on every fix of two flights; the winner cut the map error from 11.18 to 8.69 m and from 9.86 to 6.83 m, and the other three were worse on both.

Both matchers run, because neither covers the day

XFeat accepted 78% of frames at 11:38 and 0 of 40 at 15:20 on the same site with sharper imagery — the shadows had swung 100 degrees from the map's. NCC came back from git history (it is cv2.matchTemplate; nothing was installed) and correlates gradient magnitude, which does not care which side of the tree is dark. Fixes produced on 45 frames from each flight: NCC 45 and 26, XFeat 34 and 2. They also cross-check each other — corr between how far apart they are and how wrong they are is +0.928, against -0.10 for score and +0.20 for inlier count.

Overall

78%

Up from 86% claimed on 23 October — that number was withdrawn when the scoring was found to be circular. This one is scored only against the raw receiver. It flies and holds; what it is not yet is smooth, or good at every height, or good at every hour of the day.

1. What changed since the 23 October report

Everything below is in the tree now. FLOWN means it was in the air on flight 245 or later; nothing marked UNFLOWN has been tested outside the bench. The last three rows were written, flown once in a confounded build, and reverted at the pilot's instruction.

Per-track derotation	**FLOWN**	Rotation removed from every track before the fit, not from the median after it. Flight 245: flow_disagree_px p50 0.050 against 2.32, the veto firing on 2.2% of frames against 95%, velocity stale on 0.2% of rows against 41%.
Both cameras derotated	**FLOWN**	The first version did the primary only, so the gap between the two cameras became the rotation itself and the veto ate 95% of the flight. One call missing one keyword; it cost flight 163.
Tilt correction on the map fix	**FLOWN**	The matcher's answer is moved from the image centre to under the aircraft, using roll and pitch. Map error 11.18 -> 8.69 m and 9.86 -> 6.83 m. Signs measured, not derived.
NCC restored, both matchers	**FLOWN**	matcher = "both". 159 fixes on flight 245 against a previous best of 70. NCC needs a window bigger than the frame footprint or it starves — that cost 0 of 40 until it was found.
Camera yaw offset -1.95 deg	**FLOWN**	Measured on two flights (-2.03 and -1.87, spread 0.85). The sign was settled by replay; the first attempt tested the wrong one and read it as harmful.
Map fix uses its own prior	**FLOWN**	MapFuser measured the residual against the current position, so everything the estimate did while the matcher was thinking was folded in — 0.3-0.95 m against a 0.49 m correction. The pilot described this fault before it was found.
Fix rate limit, 5 m/s	**FLOWN**	A fix cannot move faster than the airframe can fly. Reported honestly: on the fixes already flown it caught neither of the two worse than 25 m.
step_dn / step_de / step_dt_s	**FLOWN**	The per-frame displacement is written down, which is what made the baseline sweep possible at all.
fix_age_s, fix_disagree_m	**FLOWN**	The matcher round trip (0.19-0.41 s) and how far the two matchers landed apart.
Deadband as a ramp	**REVERTED**	Zero at the threshold, full value at 1.5x it. Written for the jerking; reverted with the rest. Commit 5066572.
flow_vel_frames 40 -> 30	**REVERTED**	Re-swept after the derotation fix: lower phantom AND less lag, which is the rare case with no trade. Written for the toilet bowl. Commit e626f42.
fix_sigma_m 20 -> 8	**REVERTED**	The map measured 5.64 m, not the 20 m of 9 October. Commit e626f42.
Lidar correcting the barometer	**REVERTED**	A learned offset, never a replacement, so a bad lidar can never stop the aircraft flying. Commit e626f42.

2. The flight that worked — 245 / vps_00165

634 seconds on the VPS, LOITER throughout, zero ALT_HOLD. Every flight before it fell to ALT_HOLD two to four times. Scored against GPS_RAW_INT taken from the FC log, with the clock anchored on the source-switch message and checked to read 0.61 m in the first five seconds after handover.

phase	altitude	error p50	map fixes
first hold	38.7 m	0.44 m	0
climb and hold	81.8 m	1.42 m	19
holding	81.9 m	1.80 m	27
holding	82.0 m	3.40 m	17
holding	82.0 m	8.75 m	9
up and holding	107.7 m	8.96 m	11
coming down	92.6 m	3.28 m	29
last hold	32.0 m	7.20 m	0

The map halved again, and it is now the limit

flight	map fix error p50	fixes
vps_00151	11.74 m	68
vps_00154	11.18 m	70
vps_00158	9.86 m	17
vps_00160	8.24 m	6
vps_00165	5.64 m	159

by altitude on 245:		
	60-80 m	2.89 m
	80-95 m	5.49 m
	95-110 m	11.37 m

The map's closing rate is 0.192 m/s against a far slower drift, so the position settles at roughly the map's own accuracy rather than running away from it. That is what "map-limited" means, and it is the first time this project has been there. It also means the next gain comes from the map and not from the flow — and the map is four times better below 80 m than above 95.

Both matchers firing is still the best quality signal there is

Across three flights, NCC fixes split by whether XFeat also fired: both — 59 fixes, error p50 9.10 m, worst 22.90, none over 30 m. NCC alone — 73 fixes, p50 11.19, worst 83.51, 14% over 30 m. Every bad fix is one the other matcher refused. On flight 245 the split was 5.02 m against 8.61 m. Nothing else measured on this project predicts a bad fix: score manages -0.10 and inlier count +0.20.

3. What is still wrong

1. The aircraft jerks

The sent speed crosses zero 307 times in 634 seconds — 0.48 a second — and the deadband zeroes 39.9% of rows. It is a hard threshold: below it the flight controller is told exactly 0, above it the full value, so every crossing is a step input. The controller answers with tilt commands that saturate at ATC_ANGLE_MAX 15 degrees and rates to 56.9 deg/s. Map fixes are ruled out as the cause: 16% of the >1 m/s jumps land within 0.3 s of a fix, against a 15% base rate. This only became the limiting fault because the phantom finally shrank — it used to sit well above the threshold and now it sits on it.

2. Toilet bowl at 35 m

The bearing from its own centre swept 754 degrees in 24 seconds while the yaw moved 9.6 — circling, not turning. True radius p50 3.72 m, max 11.93. Period 11.4 s, and 2·pi times the 40-frame velocity baseline's 1.6 s lag is 10.1 s: the baseline's delay is the period. The first hold of the same flight, at 38 m, swept 18 degrees with a 0.78 m radius — there the deadband was zeroing 68% of the output and the loop was effectively open.

3. The barometer reads 5-7 m high

Against the lidar: 1.07x at 100 m, 1.17x at 40, 1.29x at 20, 1.72x at 10. The same climbing and descending (+6.59/+6.82/+7.00 against +6.63/+6.02/+5.24), so an offset and not the descent hysteresis remembered from the earlier airframe. At touchdown the lidar read 0.60 m and the altitude in use read 4.2. Velocity is pixels x altitude / focal, so that ratio is the velocity error — 72% at 10 m, which is the height where the aircraft was circling.

4. The map above 95 m

11.37 m against 2.89 m between 60 and 80. Fly the map work low until this is understood.

5. The error creeps

From 1.4 m to about 10 m at 82 m over a few minutes, then holds there. Consistent with the map's own 5.64 m accuracy plus the correction lag, so it may simply be what map-limited looks like — but it has not been separated.

The confound that must not be forgotten

Code was written for faults 1, 2, 3 and 5 and flown once. That flight tracked worse on the map — but the pilot had also set the origin by hand from SCR_USER1/2 instead of from the receiver on the same flight, so four things changed at once and nothing is separable. The FC log was deleted, the flight's data removed, and the tree reverted to exactly what flight 245 flew. The three commits stay in history. Reintroduce them ONE AT A TIME.

4. Closed by measurement — do not reopen

Each of these was tested on this aircraft's own data or flown and reverted. They are listed so the next attempt is a measurement and not a repeat.

Scale / focal error	0.79-1.02 against truth once both sides are smoothed equally. The "2.22x" that started the hunt was an artefact of comparing raw flow with a 1 s GPS difference.
The camera accelerometer	Never free-runs. vel_age_s p99 0.18 s against a 0.6 s timeout, zero rows over it. Flow resets the integration every 40 ms.
The gyro integration window	Bit-identical on 6400 frames. The drain window already was the frame window, because the loop blocks on the frame and the IMU queue sits behind it.
Tracker noise	98% of the per-frame step is common to both cameras, so it is real image motion. Each camera's own noise is 0.035 px against a 1.9 px step.
The derotation coefficient	Swept 14.80 / 14.30 / 13.88 / 13.40 against the hover phantom: 0.026 -> 0.024. Geometry says 13.877 and the 6.6% discrepancy costs nothing.
Aircraft oscillation	The step's energy is at 0.02-0.17 Hz; the attitude oscillates at 0.26, 0.86 and 1.03. They do not match.
The camera-to-FC lever arm	0.157 m/s at 30 deg/s against a 1.4-2.5 m/s phantom. Real, 8%, not the cause. (FC 8 cm above the CG, camera 22 cm below; pitching nose-down they move in OPPOSITE directions.)
Raising max_correction_m	Flown 3 Oct: the hold fell from 11-14 s to 4.5-9.0 s. And it is inert anyway — the Kalman gain binds first.
A height-scaled velocity baseline	Flown log 229: 11.1 m in 24 s at 31.6 m against 0.9 m on a flat baseline. An aircraft told it is slower than it is does not correct.
Declaring an honest larger sigma	Flown and reverted: 12x further at 31 m, 20x the drift rate at 17 m. This aircraft has no other horizontal source, so honesty about accuracy buys nothing.
Map window radius	Tested both ways on real frames: 30 and 40 m collapse the fix rate, 50 and 66 are identical, 90 is within noise and costs time.
Edge / corner filtering	Keeping only matches on strong structure makes it WORSE (12.11 -> 12.91 m). The matcher's own choice — flat ground — is the geometrically honest one, because a 12 m tree leans 9.7 m at the image edge and the ortho was built from different angles.
Nadir cropping	No consistent gain and the fix rate collapses.

How to measure on this project without being wrong

Four of this report's earlier conclusions were withdrawn, and all four came from the same family of mistake. Score position only against gps1_north_m / gps1_east_m, which is GPS_RAW_INT — never against gps_north_m, which is the EKF's fused output and on VPS is our own injection returning. Better still, take lat/lon from the FC .BIN, anchor the clock on the source-switch message, and check that the first seconds after handover read near zero. Filter the gps1 columns first: they contain rows up to 23 km out. Smooth both sides of any comparison equally. And before blaming a code change for a regression, replay that flight's own frames with the change on and off.